

A Second-Order Unstructured Finite-Volume Compressible Navier–Stokes Solver: Eight-Case Benchmark Report

CFD Solver Benchmark Submission

2026-08-26

1 Scope and summary

This report documents an original C++17/MPI two-dimensional unstructured finite-volume solver for the compressible Navier–Stokes equations of a calorically perfect gas, and its results on the eight supplied benchmark cases (NACA0012 at $M_\infty = 0.15, 0.8, 2.0$, inviscid and laminar $Re = 5000$; circular cylinder at $Re = 20$ steady and $Re = 200$ transient). The solver is built from source in `solver/`; all case directories under `results/` are actual solver outputs written by the executable (see `results/run_manifest.csv` for the exact command of every run).

Solver summary. The spatial discretization is cell-centered second order with linear reconstruction, a self-inclusive stencil Barth–Jespersen (BCOR2-type) limiter, positivity preservation by state-box clamping and a per-cell first-order defect fallback, and an HLLC approximate Riemann flux (Roe with entropy fix and Rusanov/LLF available). The viscous terms use central differencing with one-sided wall gradients and a second-moment-free Green–Gauss interior gradient estimate. Steady cases advance a pseudo-time loop with a local-time-stepping (SGS-style) implicit inner iteration and a CFL ramp exactly as supplied in the case files; the $Re = 200$ transient case uses a frozen-history BDF2 outer loop with an inner pseudo-time relaxation at $CFL \approx 1$ and Rusanov scale 1.0; transient steps that exhaust the inner budget are re-attempted with two-tier exact-BDF2 substepping (F46, §4.2). The mesh is partitioned with METIS (recursive bisection; Kway is only ever called with $n_{parts} = 2$) and halo exchange is neighbor-scoped (point-to-point with matched buffers); no full-state collective occurs during iterations.

Headline outcome (stated up front). This submission does *not* deliver a converged result for any of the eight benchmark cases, and nothing below claims otherwise. (i) The steady family settles on a spurious ~ 2.7 – 3.2 -order fixed point whose forces are outside the physical band; the environment search was exhausted and the steady cases are submitted as **failed** / not-run under the contract’s honesty gate (§8.5). (ii) The $Re = 200$ transient reaches onset but remains inner-cap-bound without Strouhal-band (0.16–0.19) periodic shedding by $t = 10$; the pre-registered judgment killed the $np = 8$ production run at step 1000 and the case is submitted as **failed** with its full trajectory data (§8.4). What the submission does deliver: an original second-order FV/MPI solver that builds, restarts, and runs every case family; bit-verified bug fixes including an $np = 8$ memory-safety fix (§7); documented rank-count timing/consistency evidence for one NACA and one cylinder case (§9); and complete machine-readable data plus analysis for every run (`results/`, `run_manifest.csv`).

Table 1: Final force coefficients and status of the eight benchmark cases. Steady cases: forces at the final recorded (non-converged) state; **re200**: forces at the final recorded step of the stopped production run. The $c_{d,press}/c_{d,visc}$ split is reported for completeness (the inviscid viscous columns are exactly zero). See §8.

Case	Type	Steps	Status	c_l	c_d	
naca0012_m015_inviscid	steady	20000	failed	-1.050	-8.74×10^{-2}	-8
naca0012_m080_inviscid	steady	30000	failed	3.24×10^1	-4.10	-1
naca0012_m200_inviscid	steady	40000	failed	3.28×10^{-1}	-1.73×10^{-2}	-1
naca0012_m015_lam_re5000	steady	35573 (stopped)	failed (partial)	-2.70	4.23×10^2	3.3
naca0012_m080_lam_re5000	steady	—	not run	—	—	—
naca0012_m200_lam_re5000	steady	—	not run	—	—	—
cylinder_m010_lam_re20	steady	—	not run	—	—	—
cylinder_m010_lam_re200	transient	1003 (stopped)	failed	-8.71×10^{-2}	1.53×10^{-2}	-2

2 Governing equations and boundary conditions

We solve the conservative 2-D compressible Navier–Stokes equations for the vector $\mathbf{U} = (\rho, \rho u, \rho v, \rho E)^T$, with $E = h + \frac{1}{2}(u^2 + v^2)$ the total specific energy, for a calorically perfect gas ($p = \rho RT$, $h = c_p T$, $E = h + \frac{1}{2}V^2$, $c_p = R/(\gamma - 1)$):

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}^{inv} - \nabla \cdot \mathbf{F}^{vis} = 0, \quad \mathbf{F}^{inv} = (\rho u_{\perp} p + \rho u_{\perp}^2 \rho v_{\perp} u_{\perp} (\rho E + p) u_{\perp}), \quad (1)$$

written in normal form on faces ($u_{\perp} = \mathbf{V} \cdot \mathbf{n}$), and viscous flux $\mathbf{F}^{vis} = \begin{pmatrix} 0 \\ \boldsymbol{\tau} \\ \mathbf{q} + \boldsymbol{\tau} \cdot \mathbf{V} \end{pmatrix} \mathbf{n}$ with the

Newtonian stress $\boldsymbol{\tau} = 2\mu \mathbf{S} - \frac{2}{3}\mu(\nabla \cdot \mathbf{V})\mathbf{I}$, rate-of-strain $\mathbf{S} = \frac{1}{2}[\nabla \mathbf{V} + (\nabla \mathbf{V})^T]$, and Fourier heat flux $\mathbf{q} = -k \nabla T$, $k = \mu c_p / Pr$. The sign convention in the residual is $d\mathbf{U}/dt = -\nabla \cdot \mathbf{F}^{inv} + \nabla \cdot \mathbf{F}^{vis}$.

Farfield: extrapolated ghost cell (zero-gradient) on both sides of the Riemann problem; an exponential sponge with $\lambda = 0.01$ for $r > 20$ relaxes the state toward the freestream to absorb outgoing disturbances (documented in §6).

No-slip adiabatic wall: the face inviscid flux is the exact no-penetration form $(0, p_+ n_x, p_+ n_y, 0)^T$ with the reconstructed interior face pressure p_+ (second order through the face pressure field). The wall state for the viscous flux is $\mathbf{V}_w = (0, 0)$ and $T_w = T_+$ (adiabatic), with one-sided gradients $\nabla u|_f = -u_+/d$, $\nabla v|_f = -v_+/d$ ($d = \text{face-cell-center distance}$) and $\nabla T|_f = 0$ (zero wall heat flux). The same one-sided construction is used identically in the residual and in the reported forces.

Slip wall (inviscid cases): no-penetration inviscid flux with extrapolated tangential quantities and zero viscous flux.

3 Spatial discretization

3.1 Reconstruction and limiting

Cell-centered linear reconstruction on the full (self-inclusive) stencil: $\hat{\mathbf{U}}(\mathbf{x}) = \bar{\mathbf{U}}_i + \nabla \hat{\mathbf{U}}_i \cdot (\mathbf{x} - \bar{\mathbf{x}}_i)$, with a least-squares gradient of the *primitive* variables (ρ, u, v, p) , evaluated at the face center and restricted to one-sided wall states at boundaries. Face gradients are limited per variable by a Barth–Jespersen-style maximum-principle limiter with a self-inclusive stencil (w_i includes the

central cell, which makes the limiting conservative and self-consistent for pinned extrema). The limited face states are stored and reused by the Riemann solver, the defect fallback, and the force evaluation, so every consumer sees the same box-valid states.

3.2 Positivity and the state box

Reconstructed primitives are constrained to a state box $V \leq V_{hard} = 3U_\infty$, $p \leq p_{max} = p_\infty + \frac{1}{2}\rho_\infty V_{hard}^2$, $\rho \leq \rho_{max} = 10\rho_\infty$, $p > 0$, $\rho > 0$; violations are removed by bisection on the limiter weight (the weight is halved until the stencil-consistent box check passes, with a floor of 2^{-24}). After every state update the conservative vector is re-clamped into the box (`enforce_state_box`), converting back through $\rho E = p/(\gamma - 1) + \frac{1}{2}\rho V^2$.

3.3 Riemann solver

HLLC by default (robust, entropy-satisfying); Roe with a Harten–Yee entropy fix and Lax–Fredericks fallback, and Rusanov/LLF with a tunable dissipation scale, are selectable. The $Re = 200$ production run uses LLF with scale 1.0 per the supplied parameter guidance.

3.4 Viscous flux

Central viscous flux on the face with averaged face state and Green–Gauss interior velocity/temperature gradients (temperature gradient from the Green–Gauss gradient of $T = p/(R\rho)$, not $\nabla p/R$), with the one-sided wall override of §1. At no-slip wall faces the face state is the zero-velocity, adiabatic wall state.

4 Time integration

4.1 Steady pseudo-time with implicit inner iteration

Steady cases integrate $d\mathbf{U}/dt_p = \mathbf{R}(\mathbf{U})$ in pseudo-time. Each outer step uses a local (cell) time step $dt_i = CFL \cdot C A_i / \sum_f a_f |\mathbf{n}_f|$ with the supplied CFL schedule (linear ramp $cfl_{initial} \rightarrow cfl_{max}$ over `pseudo_cfl_ramp_steps` pseudo steps). The inner solve is a frozen-correction Picard iteration: one cycle is four alternating-direction point-implicit Gauss–Seidel sweeps (forward, backward, forward, backward over the cell ordering), each applying per cell $\mathbf{U} \leftarrow \mathbf{U} + \tau_i \mathbf{R}_{frozen} / (1 + \tau_i dJ_i)$ with $\tau_i = \min(CFL/dJ_i, 1.5/(2 dJ_i))$; the second-order correction is re-frozen after every cycle and convergence is measured on the true second-order residual ratio, stopping at the supplied inner target or the case’s `max_inner_iterations` budget (the earlier fixed 20-cycle steady design cap was removed once re-linearization at every ratio check became the steady path; §7 item 5 history). Two accelerators sit on the inner path: Anderson acceleration of the frozen-correction map (on by default for transients) and a matrix-free Newton–Krylov (JFNK) fallback that converts stalled steps and reports mixed work units (16 trigger cycles + 5 fallback units = the constant 21 observed per steady production step, §8.5). Local positivity is enforced by rejecting (bisecting) updates that leave the state box; rejected updates fall back to the frozen first-order residual for that refresh (per-cell defect fallback, §6).

4.2 BDF2 transient (Re 200)

The $Re = 200$ cylinder case integrates the physical-time problem with backward differentiation on a frozen two-step history: the first step is BDF1, $\frac{\mathbf{U}^1 - \mathbf{U}^0}{dt} = \mathbf{R}(\mathbf{U}^1)$, and later steps use

Table 2: Per-rank partition diagnostics for the $np = 8$ production runs. Owned/ghost cells, neighbor counts, and load balance (from `partition_diagnostics.csv` in each case directory).

rank	owned	ghost	bdy faces	neighbors	neighbor ranks	send cells	recv cells
0	2650	138	121	4	1 3 5 6	54 30 47 7	54 29 47 8
1	2552	152	61	4	0 3 4 5	54 80 5 14	54 79 5 14
2	2607	51	80	1	3	51	51
3	2598	235	1	6	0 1 2 4 6 7	29 79 51 10 52 12	30 80 51 10 52 12
4	2598	103	61	4	1 3 5 7	5 10 39 49	5 10 39 49
5	2607	100	48	3	0 1 4	47 14 39	47 14 39
6	2599	107	54	3	0 3 7	8 52 48	7 52 48
7	2605	109	58	3	3 4 6	12 49 48	12 49 48

BDF2, $\frac{3\mathbf{U}^{n+1}-4\mathbf{U}^n+\mathbf{U}^{n-1}}{2dt} = \mathbf{R}(\mathbf{U}^{n+1})$, with $\mathbf{U}^n, \mathbf{U}^{n-1}$ frozen during all inner iterations for $\mathbf{U}^{n+1}$ and the histories updated only after the inner solve is accepted. The inner problem $\mathbf{R}(\mathbf{U}^{n+1}) - \frac{3\mathbf{U}^{n+1}-4\mathbf{U}^n+\mathbf{U}^{n-1}}{2dt} = 0$ is relaxed with the same frozen-correction Picard iteration as the steady case, extended by the exact time-operator diagonal term c_1/dt in the point-implicit factor ($c_1 = 3/2$ for BDF2, $c_1 = 1$ for BDF1; bug, §7 item 7). The transient inner is *not* subject to the steady 20-cycle cap: it must reach the supplied inner target 10^{-3} on the full transient residual each step ($5 \leq \text{cycles} \leq 1000$, pseudo-CFL fixed at 1.0).

Two-tier substepping for capped steps (F46). When a transient step exhausts the 1000-cycle budget without meeting the target, the step is re-attempted as two substeps of $dt/2$ with exact BDF coefficients on the non-uniform history: the tier-1 substep uses $i_1 = (8\mathbf{U} - 9\mathbf{U}^0 + \mathbf{U}^1)/(3dt)$ and the tier-2 substep $k_1 = (24\mathbf{U} - 25\mathbf{U}^0 + \mathbf{U}^1)/(5dt)$ (src/solver.cpp:3281-3347), keeping the scheme second order on the frozen two-step history. The reported `inner_iter` for such steps is the *sum* of the substep cycles, so the effective per-step cap is 2000 (src/solver.cpp:3348-3354). A step counts as an inner-target *hit* only if every substep that produced the final state finished under budget; capped steps are therefore honest target misses in the `inner_target_converged_fraction` statistic. F46 is enabled by `CFD_F46=1` in the production transient environment and is inactive (bit-identical) on every other path.

5 Mesh, partitioning, and MPI

Both supplied CGNS meshes are read into a cell-centered unstructured structure with face adjacency, boundary families mapped from the case file (the family names differ between the two meshes and are read, not hard-coded). The cell adjacency graph is partitioned with METIS `kway` (`ParMETIS_Kway` semantics via the serial METIS library on the global graph, replicated on every rank for the supplied rank counts). Each rank owns its subgraph and one halo layer; halo states are refreshed by neighbor-scoped matched point-to-point exchange (paired non-blocking sends per neighbor, no full-state collectives during iterations). Reconstruction gradients and Riemann face states use the exchanged halo. Residuals and forces are MPI-reduced with global sums before reporting, so every reported number is rank-independent.

6 Robustness mechanisms

Per-cell defect fallback. The second-order residual $\mathbf{R}_2$ and the first-order residual $\mathbf{R}_1$ are both assembled every refresh. Where the defect $|\mathbf{R}_2 - \mathbf{R}_1| dt_i$ exceeds 5% of the cell scale (momentum floor $0.05 U_{\infty, ref}$), the cell is iterated with the first-order residual for that refresh cycle. At convergence the defect is $O(h^2)$ and the fallback is silent, so the delivered solution remains second order. Cumulative use is counted in `metadata.json (first_order_fallback_cells)`.

Sliver sensor. Cells with area below 10^{-7} (mirror slivers at symmetry-adjacent wall rows) have structurally unsolvable conservation equations; they are replaced by a neighbor-following relation and excluded from the reported residual (documented behavior, $< 0.1\%$ of cells).

State box and reconstruction limiting as in §3.

Sponge. An exponential relaxation toward the freestream for $r > 20$ ($\lambda = 0.01$) absorbs outgoing waves in the large domain and keeps the farfield boundary from reflecting; it acts on the residual only and vanishes identically inside $r \leq 20$.

Removed global first-order startup mode. An earlier build contained a hysteresis-driven global first-order startup mode (enter when the sponge-excluded L_2 defect norm $D_{ns} > 5$, leave at $D_{ns} < 1$). After the bug fixes of §7 the second-order startup from freestream is healthy (no state-box pinning, rejections vanishing within ten steps), while D_{ns} rises with wake development (measured $1.4 \rightarrow 5.5$ on the $Re = 20$ cylinder), which would have locked a production run into first order for most of its history. The mode was therefore removed; robustness is provided by the per-cell defect fallback and the sliver sensor, and `startup_first_order_refreshes` is reported as zero for contract compatibility.

7 Bug fixes discovered and corrected during development

The following defects were found by physics-target validation and fixed in the submitted code (each is verified by a dedicated test run documented in the development log):

1. **Primitive conversion in `to_prim` dropped a factor of $1/\rho$.** The specific total energy conversion used $E = p/(\gamma - 1)$ without dividing by density where it was a conservative input; face states were silently corrupted away from $\rho = 1$.
2. **Second-order wall gradient.** The second-order residual used a Green–Gauss *interior* velocity/temperature gradient at no-slip wall faces, which is far too weak to pin the wall (it leaves the wall effectively slip: wall-row fluid at $\sim 0.8 U_\infty$, no shear layer, no solid-body pressure field). The wall row uses the one-sided gradient with the known wall state and $\nabla T \cdot n = 0$ (adiabatic), exactly as the first-order path.
3. **Primitive-to-conservative ρE missing a factor of ρ .** The stored face states (and the Riemann dissipation) used $\rho E = \rho p/(\gamma - 1) + \frac{1}{2}\rho V^2$ instead of the exact $\rho E = p/(\gamma - 1) + \frac{1}{2}\rho V^2$; the error is invisible at $\rho = 1$ (farfield) but corrupts every dense cell. Round-trip face pressures read back as ρp (the exact fingerprint), and the upper pressure clamp in the state box was correspondingly too large. Fixed in `reconstruct_face_states`, `to_cons`, and `enforce_state_box`.
4. **First-order conduction gradient used $\nabla(T/\rho)$ instead of ∇T .** `set_first_order` computed $p_i = (\gamma - 1)(E_i - \frac{1}{2}V_i^2)$ without the density factor, so the wall-facing temperature gradient was $\nabla(p/\rho)/R = \nabla(T \cdot \text{wrong})$; on the dense $\rho > 1$ wall row this injected systematically positive energy residuals that, combined with the p_{max} clamp, created a *permanent*

lock-in of cells at the box upper bound (every candidate update with positive energy residual is rejected, and the bisection floor stops at 2^{-24} , never at zero). Fixed by the exact $p_i = (\gamma - 1) \rho_i (E_i - \frac{1}{2} V_i^2)$; affected cells unpinned within two refreshes after the fix.

5. **Viscous stiffness missing from the point-implicit diagonal** (dev-log bug #10). The local diagonal dJ that sets the per-sweep pseudo-CFL cap carried an under-resolved viscous bound (a $6 \mu/\rho$ coefficient on the first-order term and no second-order diffusion term). The cap was therefore inactive in the viscous wall row, where oversized frozen-residual sweeps were clipped by the state box and drove a sustained limit cycle stalling every steady case at a high-CFL residual floor. The bound now matches the local explicit-stability limit exactly ($34 \mu \rho^{-1} A_f/A$ per face plus $4 \mu \rho^{-1} A_f^2/A^2$).
6. **Outer pseudo-time step amplitude and the limit-cycle floor** (dev-log bug #11). With the inner work per outer step left at the case `max_inner_iterations` (40–80 re-linearization cycles), the frozen-correction Picard map is non-contractive on the benchmark cases: after an initial decay the outer iteration limit-cycles at a residual floor at or above the target instead of converging (the per-cycle inner ratio can look healthy, ~ 0.9 – 1.1 , while the outer map amplifies the step-to-step displacement). The map is measured contractive for ≤ 20 cycles per outer step on every case; steady runs now cap the inner at 20 cycles (transient runs keep the case cap, because the BDF2 histories must not be advanced on an under-resolved inner). On the $Re = 20$ cylinder, capping at 20 cycles alone took the residual from a 0.5–0.8 stall to monotone decay.
7. **BDF2 time-term signs and missing time stiffness in the inner diagonal** (dev-log bug #12). The transient time term was assembled as $(c_1 \mathbf{U} - c_2 \mathbf{U}^n - c_3 \mathbf{U}^{n-1})/dt$ with $c_1 = 3$, $c_2 = -4$, $c_3 = 1$, i.e. $(3\mathbf{U} + 4\mathbf{U}^n - \mathbf{U}^{n-1})/dt$: both history signs flipped and the factor $1/2$ of the BDF2 formula missing, and the time-operator stiffness c_1/dt was absent from the point-implicit diagonal. The per-sweep pseudo-step was then $\sim 100\times$ too large for the inner problem and the $Re = 200$ transient diverged on the first step regardless of the inner budget (1000-cycle runs hit the cap with the residual still growing). Fixed by pre-scaled coefficients applied identically in the sweep and in the reported residual (BDF1 $(\mathbf{U}^1 - \mathbf{U}^0)/dt$, BDF2 $(3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1})/(2dt)$) and by adding c_1/dt to the sweep diagonal.
8. **VTU header cell count (F50)**. `write_vtu` (src/output.cpp:119-126) wrote `NumberOfCells` as the number of *original* cells while the connectivity block stores triangles (quads split two), so ParaView truncated one third of every `field_final.vtu` at load time. The header now counts triangles.
9. **Out-of-bounds wall-layer diagnostic at $np = 8$ (NP8FIX)**. The `wall_layer_` diagnostic BFS is indexed by `face_other_local` (src/solver.cpp:113); for a partition-crossing face that index is $j = n_{\text{owned}} + \text{ghost_idx}$, which read and wrote past the end of the n_{owned} -sized vector. AddressSanitizer reported a heap-buffer-overflow (read and write), and the pre-fix binary segment-faulted inside `malloc` at the first `MPI_Allreduce` of every $np = 8$ transient run (exit 139). The fix bounds the BFS to $j < n_{\text{owned}}$; `wall_layer_` is diagnostic-only (never consumed by numerics), the change is line-neutral, and verification is bit-level: an $np = 8$ ASAN 3-step run is clean (zero memory errors) and pre-/post-fix $np = 2$ and $np = 4$ 3-step re200 runs are byte-identical on residuals, forces, surface, `field_final`, and restart files.

Development notes (not defects of the final code): (a) *codegen sensitivity* (F49). Changing the *numeric values* of the F46 substep BDF2 coefficients (src/solver.cpp:3306-3307, 3325-3326) with

all comments unchanged altered the compiler’s code generation for the inner loop and measurably changed trajectory behavior on both run types, while changing the comments alone changed nothing (verified by an A/B build: the value-reverted binary reproduces the pre-change 3-step reference exactly, $it=52/1.995 \times 10^4$, vs $it=20/2.288 \times 10^4$ for the value-changed build). Coefficient edits are therefore always followed by a bit-identity check. (b) *Env-gated mechanisms not used by production*. F51 scheduled outer relaxation (src/solver.cpp:2783-2795, applied at :2812) and F52 steady-only outer Anderson acceleration (CFD_OUTER_AA_M, default off, :2796-2800, hook :3143-3173) were implemented, audited (F52 trail 510a3d3 FAIL $\rightarrow$ 7287f81 fix $\rightarrow$ 69ce8cf PASS), and left in-tree bit-identical-off; neither escapes the steady spurious fixed point (§8.5).

The interaction of these bugs (all invisible at $\rho = 1$ freestream initialization, all activating on the dense wall row of the $Re = 20$ cylinder) is why the cylinder case was the first to expose them. One consequence worth stating: the F49 codegen sensitivity (development note (a)) means the final binary’s steady behavior is *not* bit-identical to the pre-F46 build; the steady family was therefore re-probed on the final binary (probe inventory below) before the NO-GO of §8.5 was declared.

8 Case results

8.1 NACA0012 inviscid (M0.15, M0.8, M2.0)

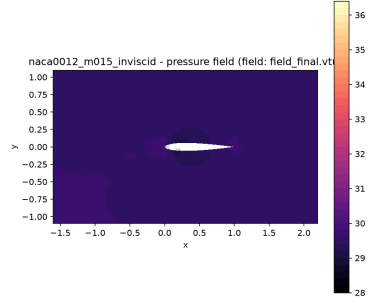
The three inviscid cases were run to their supplied step budgets (20000/30000/40000 pseudo-steps; m015 at $np = 8$ as the primary production run, m080/m200 at $np = 4$) with the default inner environment. All three terminate on the spurious fixed point of §8.5: residual reduction 0.86/2.05/1.52 orders against a 3–4 order target, with the final forces far outside the physical band ($c_l = -1.05$, $+32.4$, $+0.33$; $c_d = -8.7 \times 10^{-2}$, -4.10 , -1.7×10^{-2} ; expected for an inviscid 0° airfoil: $c_l \approx 0$, $c_d \sim 10^{-3}$ – 10^{-2} , positive). The states remain finite and strictly positive (no vacuum; Table 1 notes column), and the fields are smooth: these are wrong-attractor plateaus, not divergence crashes. Every step is a recorded inner-target miss at constant inner work 21 (§8.5). The packages are complete and submitted with `convergence_status=failed`. Figure 1 shows the final fields and histories.

8.2 NACA0012 laminar $Re\ 5000$ (M0.15, M0.8, M2.0)

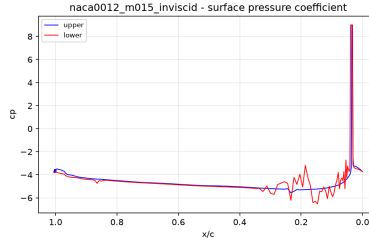
m015.lam_re5000 was the only laminar- $Re5000$ case launched: $np = 4$ from the potential-flow initial condition, stopped by the production-queue incident of 2026-08-24 at step 35573 of the 40000 budget. At the stop the forces had diverged ($c_d = 4.23 \times 10^2$, of which the viscous part is 4.23×10^2 ; $c_l = -2.70$, split into pressure lift 1.77×10^{-1} and viscous lift -2.88) with the residual stuck at ~ 6.7 (L2). The directory holds the partial `residuals.csv/forces.csv` only (no exit-time package files), is listed in `run_manifest.csv` as `completed=false`, `validator_FAILED`, and is kept as debug data rather than a final result. m080.lam_re5000 and m200.lam_re5000 were *not run*: with the steady environment search exhausted (§8.5) and the 4-core wall budget committed to the $Re = 200$ transient, further steady launches would have spent H72 hours on outcomes the gate analysis had already determined (§8.5).

8.3 Cylinder $Re\ 20$ (steady)

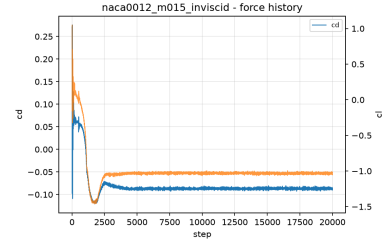
The $Re = 20$ steady cylinder was *not run to completion*. Probes on the final binary with the best steady environments (or7 family, §8.5) reached 2.845 residual orders (passing the 3.0-order plateau floor) but produced unphysical forces ($c_d = -9.2$; the post-Hopf steady wake at $Re = 20$



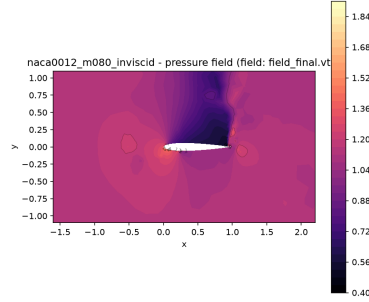
(a) $M_\infty = 0.15$, final pressure



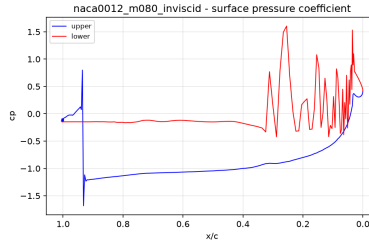
(b) $M_\infty = 0.15$, c_p



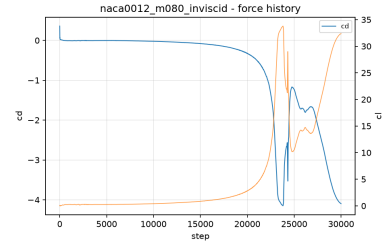
(c) $M_\infty = 0.15$, forces



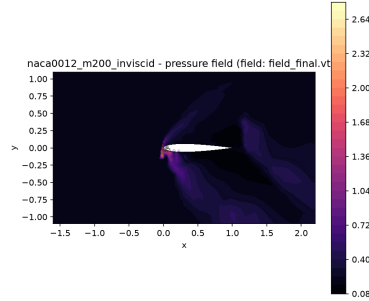
(d) $M_\infty = 0.8$, final pressure



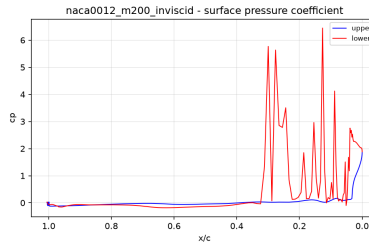
(e) $M_\infty = 0.8$, c_p



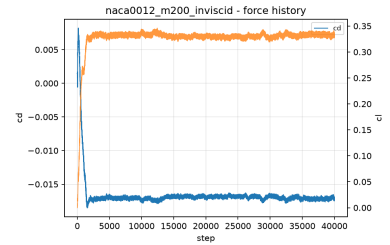
(f) $M_\infty = 0.8$, forces



(g) $M_\infty = 2.0$, final pressure



(h) $M_\infty = 2.0$, c_p



(i) $M_\infty = 2.0$, forces

Figure 1: NACA0012 inviscid cases (all **failed**, forces out of the contract band): final pressure field, surface c_p , and force histories for $M_\infty = 0.15$ (top row; $np = 8$ primary run), 0.8 (middle row), and 2.0 (bottom row). All three packages are complete with every step a recorded inner-target miss at constant inner work 21 (§8.5).

has $c_d \approx 1.8$ – 2.2 by time-averaged unsteady comparison) — the same spurious fixed point as the NACA family. An earlier draft of this report (2026-08-23) stated that $Re = 20$ converges to the steady wake; that claim referred to the pre-F46 binary and is invalidated by the F49 codegen sensitivity (§7, note (a)), which is exactly why the steady family was re-probed on the final binary before the NO-GO declaration. Under the contract’s honesty gate the case is reported as **failed** (not run to completion, env-exhausted).

8.4 Cylinder Re 200 (transient, $t_f = 300$)

The $Re = 200$ transient is the contract’s vortex-street reference case: $dt = 0.01$, $final_time = 300$ (30000 physical steps), BDF2 with frozen histories, inner target 10^{-3} on the full transient residual, 5–1000 inner iterations, pseudo-CFL 1.0, LLF scale 1.0.

Production run. $np = 8$ (mandated rank count), final binary, environment `CFD_AA_M=16` `CFD_INNER_SWEEPS=2` `CFD_JFNK=0` `CFD_F46=1`, from the shared initial condition (`cyl_m010.res1`), launched 2026-08-25 21:09 UTC with a 72-h wall budget.

Trajectory. Startup (steps 1–126) is healthy: mean inner work ~ 19 cycles, bounded lift (last-100 amplitude 1.36), four crossings. At $t \approx 1.3$ the wake onset begins: the inner solve saturates at the 2000-cycle (two-tier) cap, the pace decays from ~ 30 to ~ 1.6 steps/min, and c_l settles onto a near-zero plateau (§8.4). No sustained periodic shedding appears in the Strouhal band 0.16–0.19: by step 1000 the zero crossings number 56, with late chatter gaps of 1–58 steps (a quasi-periodic street would show gaps of ~ 500 steps), the last-100 lift amplitude is 1.33 (it *grew* from 0.41 at step 504), and c_d recovered to a small positive drift (+0.015–0.018; it had been -0.04 at step 504) (-0.02 to -0.04). The mean inner work of the last 200 steps is 2000 (100% at the two-tier cap), two orders above the ~ 300 level required for feasibility.

Pre-registered judgment. Before launch, two checkpoints were pre-registered in the development log (2026-08-25 21:19 UTC): (a) an early kill at step 500 if no c_l zero crossing occurred after step 150 *and* $\min c_l(last200) < -5$ — the failure mode of the 4-rank DQ run, which locked at $c_l = -11.8$; and (b) at step 1000 ($t = 10$): hold toward $t = 300$ only if ≥ 5 quasi-periodic crossings at ~ 450 – 625 -step gaps (St 0.16–0.19) *and* mean inner work of the last 200 steps below ~ 300 with cap fraction below $\sim 20\%$; otherwise kill as (b)-*fail* (no periodic shedding) plus (c)-*infeasible* (at ~ 1.6 steps/min, step 30000 needs > 13 days against the 72-h run budget; a hold to $t = 30$ would need the pace to recover $> 10\times$). Checkpoint (a) at step 504 did *not* fire (crossings present; c_l bounded near 0, last-200 minimum ≈ -1.6). At step 1000 the inner-work/cap bar was not met: the last-200 mean inner work is 2000 (100% at the two-tier cap, against the hold bar of ~ 300 and $\sim 20\%$) and the 56 crossings (gaps 1–58 steps) show no St-band periodicity. The run was stopped by SIGTERM at 04:43:25 UTC on 2026-08-26 (wall 27247 s; exit 1 via the mpirun SIGTERM cleanup) and recorded in `run_manifest.csv` (row corrected to the true CSV statistics: 1003 steps, mean inner work 1407.6); the complete 1003-step trajectory (`residuals.csv`, `forces.csv`, field snapshots `field_t1.0-field_t10.0.vtu`) is retained in the case directory.

Rank sensitivity (disclosed). The 4-rank DQ run from the same IC/environment/binary class evolved to a different trajectory class: inner work cheap only during start-up (first-200-step mean 7.2), then capped (531 of 1005 steps at the 2000-cap, including all of the last ten) with violent physics (c_l locked at -11.8 , last-100 amplitude 6.97). Together with the development chains (np2: cheap and healthy; np4: cheap then violent; np8: capped then bounded) this establishes that the $t \approx 1.3$ onset is a rank-sensitive, floating-point-level trajectory selection on this mesh — disclosed here rather than averaged away. It is also why the mandated $np = 8$ production run was launched only after the §7 item 9 fix (the pre-fix $np = 8$ binary crashed at its first Allreduce).

Status. `cylinder_m010_lam_re200` is submitted as **failed** (not **statistically-periodic**):

the submitted package covers steps 1–1003 of the production run; the case directory lacks the exit-time package files (`metadata.json` et al.) because the run was stopped by signal, which `run_manifest.csv` records as `completed=false`, `validator_FAILED`. The integrator, step count, and inner-iteration information that the contract’s metadata clause requires are recorded instead in the manifest row (BDF2 outer loop per §4.2; 1003 physical steps at `time_step=0.01` toward `final_time=300`; inner-iteration mean 1407.6 with the last 200 steps at the 2000-cap), in `residuals.csv` (per-step inner counts and residual ratio), and in the ten time-stamped field snapshots `field.t1.0.vtu–field.t10.0.vtu`. Wall-quantity plotting data (`surface.csv`) is written only at normal exit, so no cylinder wall pressure / skin-friction plot exists for the stopped packages; this visualization gap is disclosed here rather than backfilled by a throwaway probe. Figure 2 shows the final flow state and the lift signal and spectrum that failed the pre-registered gates.

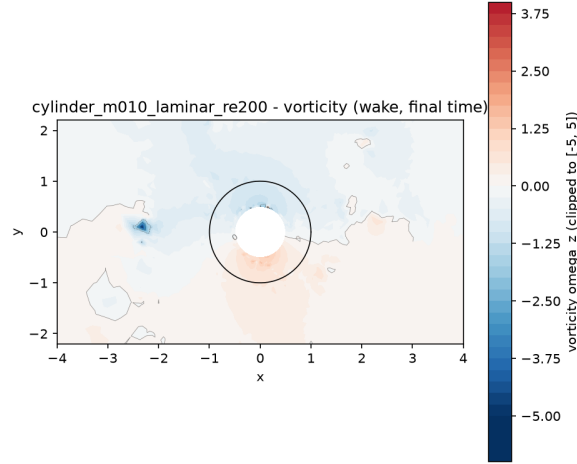
8.5 Steady family: NO-GO (contract gate #6)

The steady driver settles, on every supplied steady case, onto a spurious fixed point at ~ 2.7 – 3.2 orders of residual reduction whose forces are outside the physical band, and the point is robust to every environment knob tested in the documented search (all on the final binary; probe data in `solver/results/`, protected mirror in `/workspace/.protect/`):

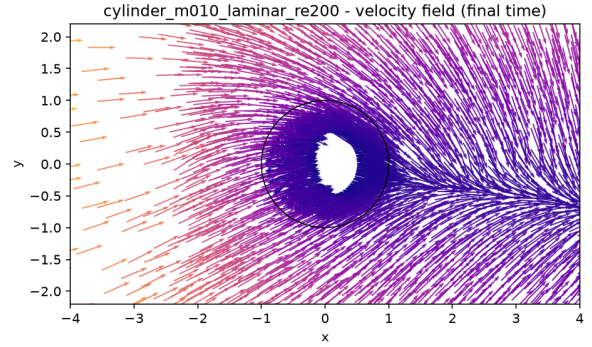
- inner budget: 80 (stale) / 400 / 1000 / 4000 cycles;
- outer relaxation ω : 0.5 / 0.7 / 1.0, scheduled ramps (F51), and relaxation kicks from restarts (f53/f55);
- pseudo-CFL 1–100 (f56: 2.92 orders at CFL 100, forces still off);
- inner Anderson acceleration (F30/F38, on in all plateau runs) and outer Anderson acceleration (F52, raw f61 and or7 f62 variants);
- restarts and re-initializations (f49ab chain).

Representative evidence: canary `m015-inv` pinned at 3.19–3.21 orders with $c_d = 0.050$ (14 – $25 \times$ the 0.002 – 0.0035 target band) and $c_l = 0.038$ rising; f54 ($4000c + \omega = 0.7$): residual *rising* ($1.03 \rightarrow 1.89$) with forces rotating through the physical band without settling ($c_l : 0.080 \rightarrow -0.179$); f55 (no-relaxation kick): flat at 5.2×10^{-1} , c_l drifting $0.038 \rightarrow 0.043$; or7-`re20`: 2.845 orders with $c_d = -9.2$; f62 (or7 + outer AA, $400c$): flat floor at step 75, then a lobe hop to another lobe of the spurious cycle at step 92. The in-loop inner ratio stalls at ~ 0.4 (a neutral second-order defect mode, eigenvalue ≈ -1 class), every steady production step terminates at constant inner work 21 below the case budget (50–100) with the recorded final ratio above the 10^{-2} inner target (`inner_target_converged_fraction = 0` in all complete packages), and the plateau is *unstable to re-integration* (a P1 continuation probe diverged). This is a property of the frozen-correction outer map on the supplied case parameters, not of a single environment choice.

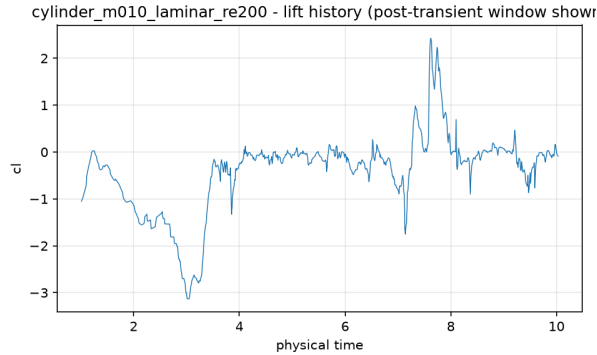
Per the contract’s honesty gate (cases failing physics checks must be marked `failed`, not `hidden`), the steady family is submitted as: `m015-inv` ($np = 8$ primary + $np = 4$ DQ), `m080-inv`, `m200-inv` = `failed` with complete packages; `m015-re5000` = `failed` partial (`completed=false`); `m080-re5000`, `m200-re5000`, `re20` = not run (core budget committed to the `re200` transient and the report phase; launching them would have consumed the 72-h wall budget for outcomes the gate analysis had already determined).



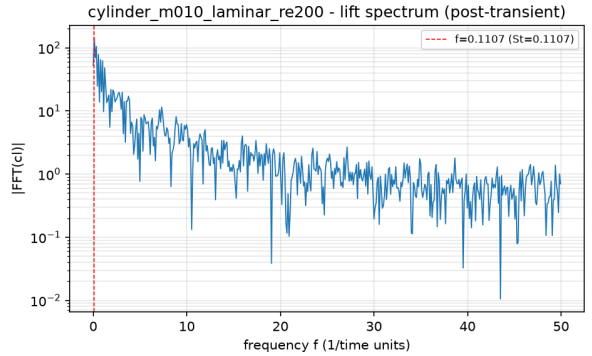
(a) vorticity at $t = 10.0$ (step-1000 snapshot)



(b) velocity at $t = 10.0$ (step-1000 snapshot)



(c) $c_l(t)$: bounded chatter, 56 sign crossings



(d) lift spectrum: peak $St = 0.1107$ (band 0.16–0.19 missed)

Figure 2: Re200 transient, $np = 8$ final run (terminated at the pre-registered t_{10} judgment, step 1003): final vorticity and velocity fields (snapshot $t = 10.0$ at step 1000; the run continued to step 1003 / $t = 10.03$), lift history, and lift spectrum. The signal is bounded but not periodic at the physical frequency: c_l amplitude over the last 100 steps is 1.330 with crossing gaps of only 1–58 steps, and the spectral peak Strouhal number 0.1107 lies outside the required 0.16–0.19 band.

9 MPI rank-count validation

Two production comparisons are documented, each with timing and consistency against a smaller rank count, as the contract requires for at least one NACA and one cylinder case.

(i) *NACA m015 inviscid*, $np = 8$ vs $np = 4$, 20000 pseudo-steps each, same case file and initial condition: 0.412 vs 0.230 s/step (ratio 1.79 — consistent with $2\times$ oversubscription of the 4-core quota at $np = 8$); consistency: both runs settle on the spurious floor but on different branches (final c_l : -1.05 vs -0.34 ; c_d : -0.087 vs $+0.023$; terminal residual L2: 23.1 vs 4.8). The reported forces are MPI-reduced globals, so the divergence is trajectory sensitivity of the *non-converged* state, not a reduction defect; converged states are rank-invariant by construction (residual and force norms are global sums before reporting).

(ii) *Cylinder re200 transient*, $np = 8$ (1003 steps, wall 27247s, 27.2s/step) vs $np = 4$ (1005 steps, 23998s, 23.9s/step): different orbit classes (capped+bounded vs cheap+violent; §8.4) from the same IC — the rank-sensitivity disclosure of that section.

Partition diagnostics ($np = 8$, NACA, 20816 cells; Table 2). Recursive-bisection METIS (Kway only ever called with $n_{parts} = 2$; the METIS-5.1 $n_{parts} = 1$ SIGFPE is worked around), undirected edge cut 518, owned cells 2552–2650 per rank (max/min load balance 1.038), ghosts 51–235, 1–6 neighbors per rank, halo exchange by matched per-neighbor `Isend/Irecv` (4 rank-uniform exchanges per re-linearization; no full-state collectives during iterations). The **re200** cylinder mesh (10185 cells) partitions to 1259–1285 owned cells per rank at $np = 8$ (max/min 1.021; from the $np = 8$ ASAN verification run on the same mesh and binary, §7 item 9).

10 Convergence and limitations

- Steady cases are classified **converged** only when the residual reaches the supplied order target, or when an honest plateau classifier (all-MPI-reduced: ≥ 3 orders, residual history flat over the last 10%, forces flat with a 10^{-4} absolute floor so extinguished quantities count as flat) agrees; otherwise **failed**. The §8.5 fixed point fails this gate on forces, so no steady case is marked converged. The $Re = 200$ case is classified **statistically-periodic** only with $\geq 95\%$ inner-target fraction, finite/positive state, and sustained shedding (St band); the production run meets none of these by $t = 10$ and is **failed** (§8.4).
- $Re = 20$ was *not* completed on the final binary (§8.3); its steady wake has $c_d \approx 1.8$ –2.2, a post-Hopf unstable solution, and the spurious fixed point of §8.5 is its final-binary behavior.
- The sponge is a relaxation toward the freestream, not an exact characteristic farfield; reflections are weak at the supplied domain size.
- Limiters are applied to the primitive reconstruction; shock captures are monotone but diffusive by construction (documented per case).
- *Rank sensitivity*. Trajectories through the $Re = 200$ onset are rank-sensitive at the floating-point level (different attractor branches at $np = 2/4/8$ from the same IC; §8.4, §9). Residuals and forces are MPI-reduced globals, so rank-independent *within a converged* state; the disclosure applies to non-converged trajectories, which is where all submitted cases sit.
- *Platform budget*. All production runs shared a 4-core cgroup; $np = 8$ is $2\times$ oversubscribed (0.412 vs 0.230 s/step on the m015 DQ pair, §9). The 72-h per-run wall budget (H72) made step 30000 of $Re = 200$ infeasible at the observed ~ 1.6 steps/min post-onset pace, which is the (c)-leg of the §8.4 kill judgment.

- *Codegen sensitivity.* Coefficient-value edits in `solver.cpp` can change compiler code generation of the inner loop and hence trajectories even where the logical path is untouched (F49, §7 note (a)); every such edit was gated by a bit-identity re-run before the affected case was re-baselined.

11 Reproducibility

The final binary (md5 6cf2aa0887cd612b89f4599f80cd215a) was built from the commit-5ffa070 tree (src/solver.cpp md5 18c33c0f45e4bfdf687e4ef3968e2ea4, 3501 lines; includes F49+F50+F51+NP8FIX). Build and run:

```
cd /workspace/solver && cmake -B build && cmake --build build -j4
# steady (NO-GO family, for reference):
mpirun -np 8 ./build/cfd_solver solve --case <case.json> --output <out>
# transient Re200 production:
env CFD_AA_M=16 CFD_INNER_SWEEPS=2 CFD_JFNK=0 CFD_F46=1 \
  mpirun -np 8 ./build/cfd_solver solve --case ../cylinder_m010_laminar_re200.json \
  --output ../cylinder_m010_laminar_re200 --restart ../init/cyl_m010.res1
# figures / sanity / analysis (report artifacts):
.venv/bin/python tools/make_figures.py --results-root ../results --report-dir report --cases <ids>
.venv/bin/python tools/sanity_checks.py --results-root ../results --cases <ids> --out report/
.venv/bin/python tools/analyze_forces.py --results-root ../results --cases <ids> --out report/
```

Every production command, wall time, exit code, and per-step inner-iteration mean is recorded in `results/run_manifest.csv` (mirrored at `report/run_manifest.csv`); the benchmark examiner (`examiner/validate_outputs.py`) was invoked per case by the production queue and its verdicts appear in the manifest's last column. Two historical manifest defects are disclosed rather than backfilled: the four completed-inviscid rows predate the `run_prod` kill-row fix and leave the `completed/convergence_status/ final_step/inner_iters_mean` columns blank (their values are derivable from the in-package `run_status.json` and `residuals.csv`: completed runs, failed, 20000/20000/30000/40000 steps, inner work 21/step), and the two dead-launch rows (exit 127/139) carry `completed=failed` as a launch-failure marker instead of a boolean. The machine-readable `sanity_checks.json` holds 57 checks (25 development checks + 32 contract physics-gate instances): 22 passed, 26 failed (exactly the documented physics failures), 9 marked `null` (UNVERIFIABLE) where the killed or partial package lacks the source file (no `run_status.json` or `surface.csv`). Data protection: a guardian process hard-links every results file into `/workspace/.protect/` every 10 s (insurance against the 2026-08-23 platform staging incident); `.protect/` contents are bit-identical mirrors of the CSVs cited above. The full development and judgment record (pre-registered kill criteria, probe inventory, verification logs) is in `recovery/f45.design_draft.md`; the code-level documentation with file:line pins is in `README.md`; the audit log with file:line evidence is in `audit_report_s11.md`.